

Ibrahim Ahmed Ibrahim

SOFTWARE ENGINEER · BACKEND ENGINEER & SYSTEMS ARCHITECTURE

Egypt

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Summary

Software Engineer specializing in Backend Systems Architecture and AI Agent Integration. Versatile across multiple technology stacks — including FastAPI, ASP.NET Core, and Node.js with hands-on experience delivering full-stack products from API design to production deployment. Proven record of integrating LLMs and RAG pipelines.

Skills

Languages	C#, Python, TypeScript/JavaScript, SQL, Bash, LaTeX
Backend	ASP.NET Core, FastAPI, EF Core, LINQ, Pydantic
Frontend	React, Vanilla JS, HTML/CSS
Databases	MS SQL Server, PostgreSQL, MongoDB, Redis, Pinecone (Vector DB)
AI & Agents	Semantic Kernel, RAG Pipelines, LangChain, LLM Orchestration, Context Engineering, Ollama, LM Studio, VLLM
DevOps & Tools	Docker, RabbitMQ, Swagger/OpenAPI, Git, Postman, Socket.io
Engineering	Clean Architecture, System Design, SOLID, Design Patterns, REST APIs, Agile, OOP, Caching Strategies, Data Structures & Algorithms, Microservices, Event-Driven Architecture

Work Experience

Techno Future

Full-Stack Software Engineer

Dec 2025 – Present

- **Techno Terminal:** Initiated and engineered the academy's internal full-stack management platform (students, academics, finance, competitions, HR, notifications, tasks, analytics) — a modular FastAPI + PostgreSQL monolith across **11** business modules with Router → Service → Repository layering, D+ hybrid slices, UoW and stateless DI patterns, strict Pydantic DTOs, and Supabase JWT role-based access.
- **Backend & API:** Built a FastAPI REST API with **29** routers across academics, CRM, finance, HR, competitions, notifications, tasks, and analytics; standardized `{success, data, error}` response envelope with exception → HTTP mapping; integrated Twilio WhatsApp and Gmail notification service.
- **Database:** Designed a PostgreSQL schema of **33** tables and **22** analytical views across 17 modular schema files; **87** incremental SQL migrations for production upgrades, with versioned indexes, triggers, functions, and constraints.
- **Quality & Testing:** **840** automated tests across **42** files covering all **11** modules, using transaction-isolated fixtures and mocked JWT auth, with a GitHub Actions CI pipeline.
- **Frontend & Deployment:** React 19 + TypeScript + Vite SPA with TanStack Query, Zustand, Axios JWT interceptor, Tailwind CSS, and Recharts dashboards; frontend on Vercel, FastAPI backend on Leapcell (Gunicorn/Uvicorn) with health-check endpoints and environment-driven config.

Techno Future

Software Instructor (promoted to Head of Instructors)

Mar 2025 – Dec 2025

- Designed and delivered comprehensive educational roadmaps for different technology tracks: Database/SQLite, Python, Frontend (Vanilla JS), and Backend/APIs with Express.js.
- Trained **50+** students with a **95%** completion rate; mentored student teams to achieve **top 4 positions** in the Afro-Asian Competition for Technology & Innovation.
- Promoted to Head of Instructors within **9 months** based on demonstrated leadership and technical performance.
- Led strategic planning and curriculum roadmaps across **5+** technology tracks, managing a multidisciplinary team of **5+** instructors.
- Conducted regular code reviews and technical audits across concurrent student/instructor projects, enforcing quality standards and engineering best practices.
- Led technical recruitment: defined role requirements, designed assessments, and conducted interviews to expand the instructional team.

Outlier

Remote

Prompt Engineer, Responses Evaluation

Oct 2024 – June 2025 · 8 mos

- Worked as a remote contractor at Outlier.ai, evaluating AI responses, developing testing methodologies, and collaborating on model improvement.
- Analyzed behaviors and created detailed reports to enhance model performance.
- Designed challenging prompts to stress-test models in coding and software engineering domains.
- Contributed to refining algorithms for generative models to better serve coding-related tasks.
- **Skills:** LLM Evaluation & Reporting, Prompt Engineering, Technical Writing & Documentation, Remote Collaboration, Technical Analysis & Troubleshooting.

Freelancer

Remote

.NET Backend Engineer

2024 – May 2025

- Developed scalable ASP.NET Core Web APIs for diverse clients, implementing Clean Architecture and Repository patterns for maintainability and testability.
- Optimized database performance using Entity Framework Core and LINQ with strategic indexing and caching, reducing query execution times significantly.
- Built secure RESTful APIs with JWT authentication, handling daily transactional workloads for client applications.
- **Technical Skills:** C#, ASP.NET Core, Entity Framework Core, SQL Server, REST API, Docker, Git, Clean Architecture.

Projects

Techno Terminal — [GitHub](#) → | [Live](#) →

Techno Future Academy

comprehensive CRM & operations platform, managing student lifecycles, financial transactions, and competition tracking

Feb 2026 – Present

- **Architecture:** Modular Monolith (API → Module[Services → Repository] → DB) across 10 domain modules; enforced Repository Pattern, Dependency Injection via FastAPI, strict Pydantic DTOs, Factory + Strategy patterns for per-request service instantiation.
- **Database:** Designed PostgreSQL schema (core & history/tracking tables) with analytical views, partial unique indexes for enrollment integrity, and a hybrid migration workflow: canonical `schema.sql` + 21 incremental SQL migrations + Alembic revision baseline.
- **Auth & Transport:** Integrated Supabase JWT with **HTTPBearer** guards and local UID mapping for role-based access (admin / system_admin). FastAPI REST API serves external web/mobile clients — eliminating business logic duplication without extra HTTP hops.
- **Quality & Deployment:** Achieved **94%** test coverage (161 tests across 20 modules) using transaction-isolated fixtures and JWT mocking. Deployed via Gunicorn + Uvicorn workers with `/health` liveness endpoint and environment-driven configuration.
- **Frontend & Analytics:** React 18 + TypeScript + Vite frontend with TanStack Query, Zustand auth store, and Axios JWT interceptor. Implemented 4 analytics sub-routers (academic, financial, competition, BI) using PostgreSQL window functions and aggregation views for real-time dashboards.

OllamaNet — [GitHub](#) →

Modular LLM platform built with .NET microservices, enabling users to converse with, manage, and explore LLMs via independently deployable services with Clean Architecture.

May 2025

- **Architecture:** Designed modular microservices using ASP.NET Core with Clean Architecture, Redis distributed cache, RabbitMQ messaging, and Ocelot API Gateway.
- **.NET Patterns:** Implemented Repository pattern, dependency injection, Polly resilience policies, and FluentValidation for robust error handling.
- **Features:** Built AuthService (JWT), ConversationService (real-time chat), ExploreService (model discovery), and AdminService with Server-Sent Events for streaming.
- **Technical Skills:** .NET 9, ASP.NET Core, EF Core, SQL Server, Redis, RabbitMQ, Ocelot, Swagger, Semantic Kernel, Docker.

ConversationService — [GitHub](#) →

Core .NET microservice managing AI-powered conversations with real-time chat, message history, folder organization, and RAG document processing.

May 2025

- **Architecture:** Implemented Clean Architecture with ASP.NET Core, separating domain logic, application services, and infrastructure layers.
- **.NET Features:** Used EF Core for data persistence, Redis for caching, Polly for retry policies, and background services for async processing.
- **Performance:** Achieved response time <100ms (cached), 99.9% uptime, and cache hit ratio >80% through strategic caching.
- **Technical Skills:** ASP.NET Core, EF Core, Redis, Pinecone, Semantic Kernel, RabbitMQ, Polly, FluentValidation, LINQ.

NeoStreamJS — [GitHub](#) →

Architected a lightweight Node.js framework for rapid interactive web app development, enabling developers to build UI-powered tools with just script-like JavaScript.

May 2025

- **Architecture:** Engineered a zero-build system using Node.js, Express.js, and WebSockets for real-time bidirectional state synchronization.
- **Impact:** Eliminated frontend build steps, reducing project setup time by **90%** and simplifying internal tool development.
- **Technical Skills:** Node.js, Express.js, WebSockets, Vanilla JavaScript, HTML/CSS, Event-Driven Architecture.

LLM-Inference — [GitHub](#) →

Engineered a distributed inference system allowing users to deploy and expose local Ollama LLM models as public APIs via ngrok and RabbitMQ.

May 2025

- **Architecture:** Orchestrated a notebook-first deployment pipeline integrating Ollama (local LLM), ngrok (tunneling), and RabbitMQ (discovery).
- **Features:** Enabled seamless remote access to local models, supporting multi-model deployment and automated service registration.
- **Performance:** Successfully handled concurrent requests with reliable service discovery and low-latency tunneling.
- **Technical Skills:** Python, Jupyter Notebook, Ollama, ngrok, RabbitMQ, Uvicorn, pika, requests.

Education

Kafr El-Sheikh University

Faculty of Computer Science

- Computer Science major
- **Courses:** Data Structures & Algorithms, Project Management, Software Engineering, Web Development, Information Systems, AI & Machine Learning, Computer Networks

Kafr El-Sheikh, EG

2021 - 2025

DEPI

Digital Egypt Pioneers Initiative

- Web Development .Net Track
- **Courses:** Web development, Database Managing, MS SQL Server, Advanced C#, LinQ, Entity Framework, REST API, MVC, business English, FreeLancing

Hybrid

May 2024

Publications

- **Starting With Ollama:** Run and Customize Large Language Models (LLMs) Locally (September 2024) — **Medium**
- **Introduction to Ollama's LLMs:** Basics, Parameters, Templates, and Configurations (October 2024) — **Medium**
- **Code First vs. Database First in EF Core:** Which Approach Should You Choose? (October 2024) — **Medium**